ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MATHEMATICAL LITERACY

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mathematical Literacy practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct methods and well-reasoned advice.

QUESTION 1

1.1.1 $850 \times R10{,}84$ ✓ $= R9\ 214{,}00$ (20 years) ✓; $850 \times R10{,}09 = R8\ 576{,}50$ (30 years) ✓ (3)

1.1.2 $9\ 214 \times 240$ ✓ $= R2\ 211\ 360$ ✓; $8\ 576{,}50 \times 360 = R3\ 087\ 540$ ✓ (3)

1.1.3 R876 180 more ✓; choose 20 years if they can afford the higher payment, because it saves a lot of interest ✓ (2)

1.2.1 Option 1: $20\ 000(1 + 0{,}075 \div 12)^{60}$ ✓ $= R29\ 065{,}89$ ✓; Option 2: $20\ 000(1 + 0{,}08 \times 5)$ ✓ $= R28\ 000$ ✓ (4)

1.2.2 Option 1 ✓, by R1 065,89 ✓ (2)

1.3.1 $15\ 000 \times 1{,}04$ ✓ $= R15\ 600$ ✓ (2)

1.3.2 $15\ 000 \times 1{,}06$ ✓ $= R15\ 900$ ✓ (2)

1.3.3 Worse off ✓ – prices rose more than her salary, so she can buy less than before (about R300 less buying power per month) ✓ (2)

1.4.1 $48\ 500 - 21\ 300$ ✓ $= R27\ 200$ ✓ (2)

1.4.2 Expenses $= 6\ 000 + 12\ 000 + 2\ 450 = R20\ 450$ ✓; $27\ 200 - 20\ 450$ ✓ $= R6\ 750$ ✓ (3)

1.4.3 $\frac{6\ 750}{48\ 500} \times 100$ ✓✓ $= 13{,}92\%$ ✓ (3)

1.5 $420\ 000 - 370\ 500 = 49\ 500$ ✓; $0{,}31 \times 49\ 500 = 15\ 345$ ✓; $77\ 362 + 15\ 345 = R92\ 707$ ✓; $92\ 707 - 17\ 235$ ✓ $= R75\ 472$ per year ✓; $\div 12$ ✓ $= R6\ 289{,}33$ per month ✓ (7)

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QUESTION 2

2.1.1 $6 \times 5 = 30\ m^2$ ✓; $2 \times 2 = 4\ m^2$ ✓; $30 - 4 = 26\ m^2$ ✓ (3)

2.1.2 $26 \times 1{,}1 = 28{,}6\ m^2$ ✓; tile $= 0{,}6 \times 0{,}6 = 0{,}36\ m^2$ ✓; $28{,}6 \div 0{,}36 = 79{,}44$ ✓ → 80 tiles ✓ (4)

2.1.3 $80 \div 4 = 20$ boxes ✓; $20 \times R289$ ✓ $= R5\ 780$ ✓ (3)

2.2.1 $8 \times 4 \times 1{,}5 = 48\ m^3$ ✓✓ $= 48\ 000\ \ell$ ✓ (3)

2.2.2 $48\ 000 \div 25 = 1\ 920$ minutes ✓✓ $= 32$ hours ✓ (3)

2.2.3 $48\ 000\ \ell = 48\ k\ell$ ✓✓; $48 \times R28{,}40$ ✓ $= R1\ 363{,}20$ ✓ (4)

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QUESTION 3

3.1.1 $\text{A: } 312 \div 100 = 3{,}12 \text{ h } ✓ = 3 \text{ h } 7 \text{ min } ✓; \text{ B: } 268 \div 75 = 3{,}57 \text{ h } ✓ = 3 \text{ h } 34 \text{ min } ✓$ (4)

3.1.2 $\text{A: } 312 \times 8 \div 100 = 24{,}96\ \ell\ ✓; 24{,}96 \times 23{,}45\ ✓ = \text{R}585{,}31\ ✓; \text{ B: } 268 \times 8 \div 100 = 21{,}44\ \ell\ ✓; 21{,}44 \times 23{,}45\ ✓ = \text{R}502{,}77\ ✓$ (6)

3.1.3 $502{,}77 + 62\ ✓ = \text{R}564{,}77\ ✓$ (2)

3.1.4 Either route with TWO valid reasons ✓✓✓, e.g. Route A – about 27 minutes faster and easier, safer driving; or Route B – about R20,54 cheaper and shorter distance (3)

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QUESTION 4

4.1.1 Class A's median (61%) is higher than Class B's (55%) ✓; Class A's pass rate (78%) is higher than Class B's (64%) ✓; Class B's mean may be pushed up by a few very high marks ✓, as its range is much larger ✓ (4)

4.1.2 Class A ✓ – its range (45%) is smaller, so the marks are closer together ✓ (2)

4.1.3 $78\% \times 32\ ✓ = 24{,}96 \approx 25 \text{ students } ✓$ (2)

4.1.4 A few students scored very high marks (outliers) ✓, which raises the mean but not the median ✓ (2)

4.2.1 $\frac{1\,610 - 1\,250}{1\,250} \times 100\ ✓✓ = 28{,}8\%\ ✓$ (3)

4.2.2 A line graph ✓ – it shows the trend over time / a bar graph (accept with the reason that it compares separate years) ✓ (2)

4.3 Any THREE: the axis does not start at zero ✓; the sample is small or biased ✓; only the averages that support the claim are given ✓; percentages without actual numbers; data taken out of context (3)

4.4 A small sample may by chance not represent the whole population ✓; a larger sample gives results closer to the truth / more reliable ✓ (2)

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QUESTION 5

5.1.1 $\frac{1}{500}\ ✓$ (1)

5.1.2 $\text{Expected winnings} = \frac{1}{500} \times 5\,000 = \text{R}10\ ✓; \text{expected value} = 10 - 20 = -\text{R}10\ ✓; \text{on average a person loses R}10 \text{ per ticket bought } ✓$ (3)

5.2 $10 \times 10 \times 10 \times 10 = 10\,000 \text{ possible PINs } ✓; \text{P} = \frac{1}{10\,000}\ ✓$ (2)

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5.3 Insurance costs $12 \times 89 = \text{R}1\,068$ per year ✓; expected loss $= \frac{1}{25} \times 6\,000$ $= \text{R}240$ per year ✓; on average the insurance costs more than the expected loss ✓; but a student who cannot afford to replace the phone may still choose insurance for peace of mind (any sensible advice) ✓ (4)

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TOTAL: 100

END OF MARKING GUIDELINE